Sterling McKinley

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PROFESSIONAL SUMMARY

Engineer with extensive experience managing high-availability, mission-critical systems across enterprise hybrid and AWS cloud environments. Proven track record of bridging application support, production triage, deep root cause analysis (RCA), and incident response to rapidly drive down MTTR. Strong technical foundation in Linux, Kubernetes, and containerization paired with experience in observability, Infrastructure as Code (IaC), and disaster recovery. Passionate about eliminating operational toil through automation (Python/Bash), CI/CD pipelines, and proactive SLO-driven telemetry.

TECHNICAL SKILLS

Cloud & Infrastructure: AWS EC2, VPC, S3, RDS, Infrastructure as Code Terraform, Ansible

Site Reliability: Root Cause Analysis, Error Budgets, SLI/SLO/SLA, MTTA/MTTD/MTTR, Disaster Recovery, ITIL Framework (Incident, Change, Problem Management)

Monitoring/Observability: Datadog, New Relic, Dynatrace, Prometheus, Grafana, AWS CloudWatch, Observe

Containers & Orchestration: Kubernetes, EKS, ECS, Docker

CI/CD & Version Control: GitHub, GitHub Actions, GitLab, Jenkins, Bitbucket, IBM UrbanCode Deploy

Operating Systems & Middleware: Linux (RHEL, Ubuntu), Windows Server IIS, AIX, Solaris, Tomcat, JBoss, WebSphere

Programming Languages & Scripting: Python, Bash

Automation/AI: GitHub Copilot, Anthropic Claude, Google Gemini, Windsurf

Tools: Visual Studio, Confluence, Jira, PagerDuty, ServiceNow, G Suite, Microsoft Office Suite, Microsoft Teams, Slack, Zoom, Postman

PROFESSIONAL EXPERIENCE

Capital One | Chicago, Illinois

Platform Engineer / Site Reliability Engineer | 01/2023 - Present

- Champion operational readiness, cloud-native resiliency, and continuous compliance across an enterprise portfolio of 500+ microservices and applications within Intelligent Foundations & Experiences (IFX).
- Spearheaded SLI/SLO definition and telemetry standards across critical AWS workloads; engineered multi-window burn-rate alerts that yielded a 93% SLO attainment rate and reduced critical production incidents by 30%.
- Modernized observability platforms across New Relic, Observe, and CloudWatch; facilitated blameless postmortems and systematic root-cause analyses (RCA) that cut service interruption durations by 20%.
- Engineered automated vulnerability scanning workflows across GitHub repositories using custom Python/Bash scripts and AI models to parse issue backlogs, categorize risk severity, and deliver actionable remediation playbooks to engineering leads.
- Provisioned and standardized scalable AWS cloud infrastructure (EKS, ECS, EC2, Lambda, RDS) using Terraform modules to ensure secure, reproducible, multi-AZ production environments.
- Partnered with application engineering teams to conduct comprehensive Production Readiness Reviews (PRRs), Application Recovery Plans, architecture diagram evaluations, and resilience/chaos testing to enforce strict governance, risk, and compliance (GRC) standards across enterprise workloads.
- Spearheaded cross-functional disaster recovery (DR) simulations and failover exercises, validating automated recovery mechanisms to satisfy annual corporate resiliency compliance benchmarks.
- Standardized ITSM governance by utilizing ServiceNow to manage end-to-end implementation and configuration change requests, adhering strictly to ITIL standards to minimize release risk across critical cloud workloads.
- Built Python and Bash self-service automation workflow scripts to remove deployment bottlenecks, streamline developer experience, and maintain zero-downtime CI/CD releases.
- Designed and deployed centralized executive dashboards to provide real-time visibility into security postures and business-critical delivery milestones, accelerating cross-functional remediation cycles and risk tracking.
- Integrated AI engineering tools (GitHub Copilot, Gemini, Windsurf) into operational workflows to automate telemetry extraction, log parsing, and recurring maintenance tasks.

PNC | Remote

Infrastructure Configuration Consultant | 07/2019 - 01/2023

- Served as Tier-3 escalation lead for 24/7/365 production operations across enterprise core banking platforms, troubleshooting high-severity incidents and driving rapid MTTR.
- Engineered, patched, and maintained enterprise middleware and container environments (OpenShift, WebSphere, JBoss) on high-availability RHEL and Windows Server platforms.
- Executed deep-dive root cause analysis by diagnosing application thread/heap dumps, OS bottlenecks, and network anomalies to prevent recurring system failures.
- Leveraged ITIL frameworks within ServiceNow to author and process implementation and configuration Change Requests (CRs), ensuring zero-downtime production deployments and continuous compliance.
- Managed end-to-end release delivery pipelines using IBM UrbanCode Deploy and Bitbucket, automating configuration scripts and cutting deployment failure rates by 20%.
- Developed automated health-check scripts and proactive synthetic monitoring to detect system degradation before business impact.
- Monitored applications and systems to track performance, proactively troubleshoot, and automate system management to prevent disruptions.

Peraton (formerly Perspecta) | Remote

Application Integrator | 07/2018 - 07/2019

- Provided technical support and platform administration for the Federal Reserve Bank of New York Auction Reporting (TAAPS) critical financial application.
- Deployed, configured, and monitored multi-tier enterprise web application infrastructure on IBM WebSphere Application Server ND and IBM HTTP Server.
- Managed full-lifecycle implementation and configuration change requests via ServiceNow following ITIL protocols, streamlining approvals and mitigating production disruption.
- Collaborated with security and compliance teams to enforce stringent financial, SOC 2, and regulatory security standards across underlying host systems.

Hyatt | Chicago, IL

Operations Analyst | 11/2017 - 07/2018

- First point of contact for Hyatt's global World of Hyatt LMS (Loyalty Management System) related inquiries regarding operational processes and issues.
- Provided support for third-party vendor software, LMS code deployments, and maintenance releases, and functioned as team liaison working with hotel locations, application development teams, corporations, partnerships, and external vendors to triage LMS issues
- Created and maintained documentation for solution requirements, use cases, user acceptance criteria, and training material

Hansen Technologies (formerly Sigma Systems) | Remote

Technical Account Manager | 03/2017 - 11/2017

- SaaS (Software as a Service)
- Served as primary technical contact for the client in Caribbean markets (Miami, Barbados, Bahamas, Jamaica, Trinidad, St. Lucia, St. Vincent, Grenada, India)
- Managed and supported Sigma Systems' proprietary software
- Installed, configured, administered, and supported Java applications/systems hosted on Unix/Linux operating systems utilizing middleware technology (Oracle WebLogic Server and Red Hat JBoss)
- Debug Java applications in production environments, including stack traces, logs, and exception analysis
- Monitored applications and systems to track performance, proactively troubleshoot, and automate system management to prevent disruptions.

HCSC – Blue Cross Blue Shield | Remote

Sr. Software Systems Programming Specialist | 07/2011 - 03/2017

- On call Support (1st shift & 3rd shift)
- Installed and configured IBM WebSphere Application Server in vertical and horizontal clustered environments
- Migrated applications hosted on WebSphere version 6.x to WebSphere version 8.5.x and WebSphere version 7.x to WebSphere version 8.5
- Tasked with deployments, administration, configuration, upgrades, maintenance, and troubleshooting of J2EE applications in WAR, JAR & EAR formats on WebSphere Application Server in multi-clustered, high-availability environments
- Defined and configured application servers, JVMs, Virtual Hosts, Web Applications, Web Resources, Servlets, MQ, JDBC providers, and data sources connecting to various databases, including SQL, DB2, and mainframe
- Debug Java applications in production environments, including stack traces, logs, and exception analysis

- Monitored applications and systems to track performance, proactively troubleshoot, and automate system management to prevent disruptions.

RELEVANT PROJECTS

sre-3tier-terraform: Designed a highly available, multi-AZ AWS architecture provisioned entirely through Terraform. Conducted chaos engineering experiments using AWS Fault Injection Simulator (FIS) to validate auto-healing and disaster recovery mechanisms under simulated outage conditions.

multi-tier cloud-native observability – Architected an end-to-end containerized telemetry pipeline utilizing Docker Compose, Prometheus, and production Grafana dashboards to monitor distributed services with real-time alerting.

ai-chatbot-with-amazon-bedrock – Engineered a conversational AI chatbot using Python and the Amazon Bedrock Converse API with custom Guardrails to automate internal operations and incident runbook lookups.

game-day-notifications – Developed an event-driven AWS serverless architecture leveraging Python, AWS Lambda, EventBridge, and SNS to dispatch automated real-time SMS/Email status updates.

CERTIFICATIONS

Google AI Professional Certificate – Google

AWS Certified AI Practitioner (AIF-C01) – Amazon Web Services

Reliability Engineering: Measuring and Managing Reliability – Google

Google AI Essentials – Google

KCNA: Kubernetes and Cloud Native Associate – The Linux Foundation

AWS Certified Cloud Practitioner – Amazon Web Services

EDUCATION

Eastern Illinois University | Charleston, Illinois
Bachelor of Science in Management Information Systems

Kura Labs | Virtual
Cloud /Site Reliability Engineering – Apprenticeship | 07/2022 - 12/2022